Do not write outside the box

**0 1** This question is about the extraction of aluminium.

**0 1 . 1** An aluminium atom is represented as:

$$^{27}_{13}Al$$

Give the number of electrons and neutrons in the aluminium atom.

**[2 marks]**

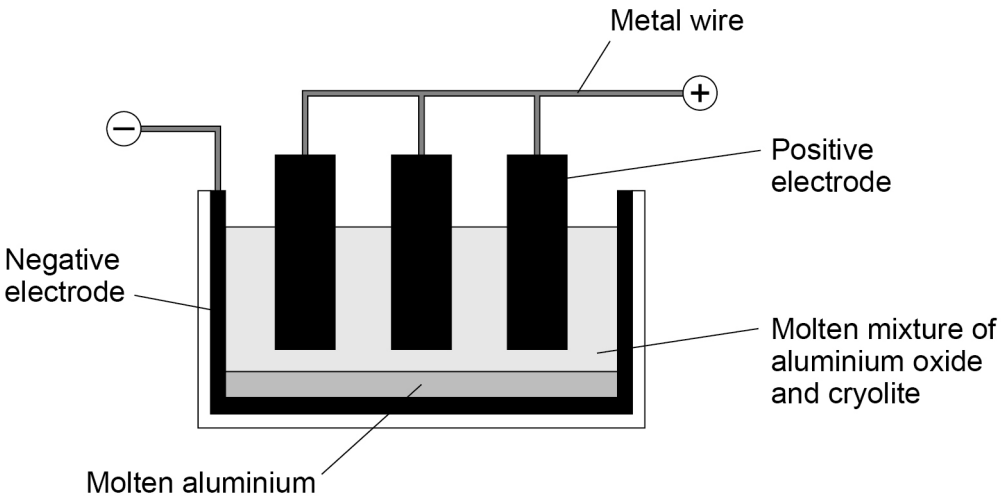
Number of electrons _____

Number of neutrons _____

Aluminium is extracted by the electrolysis of a molten mixture of aluminium oxide and cryolite.

**Figure 1** shows the cell used for the electrolysis.

**Figure 1**




**0 1 . 2** Aluminium is produced by the reduction of aluminium oxide ($Al_2O_3$).

What is meant by the term reduction?

**[1 mark]**

_____

_____

Do not write outside the box

**0 1 . 3** Oxygen is formed at the positive carbon electrodes.

Explain why the positive carbon electrodes must be continually replaced.

**[3 marks]**

_____

_____

_____

_____

_____

_____

**0 1 . 4** A substance conducts electricity because of free moving, charged particles.

What are the free moving, charged particles in a:

- carbon electrode (made from graphite)
- molten mixture of aluminium oxide and cryolite
- metal wire?

**[3 marks]**

Carbon electrode (made from graphite) _____

Molten mixture of aluminium oxide and cryolite _____

Metal wire _____

9

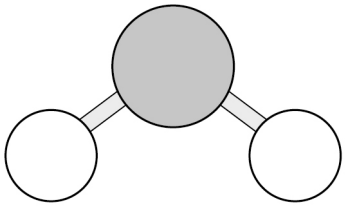
**Turn over for the next question**

Do not write outside the box

**0 2** This question is about substances with covalent bonding.

**0 2 . 1** **Figure 2** shows a ball and stick model of a water molecule ($H_2O$).

**Figure 2**



Suggest **one** limitation of using a ball and stick model for a water molecule.

**[1 mark]**

_____

_____

_____

**0 2 . 2** Ice has a low melting point.

Water molecules in ice are held together by intermolecular forces.

Complete the sentence.

**[1 mark]**

Ice has a low melting point because the

intermolecular forces are _____.